SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0307	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE-AI	**Date:**	

Code of Conduct

*I __**A.Sai Rohit (192472144)**__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: __**A.Sai Rohit**__

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1.Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

a) Referring Expressions (Mentions) and Possible Antecedents

The given paragraph is: "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home." Before resolution can be attempted, every noun phrase and pronoun in the discourse (called a 'mention') must be identified, since coreference resolution is fundamentally the task of partitioning the set of mentions into equivalence classes that refer to the same real-world entity.

Mention (Referring Expression)	Type	Sentence No.	Possible Antecedent(s)
John	Proper noun (Discourse-new entity)	1	— (introduces entity E1)
Mary	Proper noun (Discourse-new entity)	1	— (introduces entity E2)
the park	Definite NP (Discourse-new entity)	1	— (introduces entity E3, a location)
He	Pronoun (3rd person, masc., singular)	2	John (E1)
a ball	Indefinite NP (Discourse-new entity)	2	— (introduces entity E4)
She	Pronoun (3rd person, fem., singular)	3	Mary (E2)
it	Pronoun (3rd person, neuter, singular)	3	a ball (E4) / possibly 'the park' (E3)

The dog	Definite NP — note: not previously introduced ('bridging' mention)	4	— (introduces entity E5)
him	Pronoun (3rd person, masc., singular)	4	John (E1)
they	Pronoun (3rd person, plural)	5	John + Mary (E1+E2) / possibly + dog (E1+E2+E5)
home	Common noun (Discourse-new / deictic location)	5	— (introduces entity E6)

Six discourse entities are therefore introduced: E1 = John, E2 = Mary, E3 = the park, E4 = a ball, E5 = the dog, and E6 = home. The pronouns 'He', 'She', 'it', 'him' and 'they' are the anaphors that require resolution, each one searching backward through the discourse for a licensed antecedent from among E1–E6.

b) Applying the Resolution Constraints

Each of the four stated constraints is applied systematically as a filter over the candidate-antecedent set of every pronoun. The constraints are applied in the order (1) Gender/Number Agreement → (2) Semantic Compatibility → (3) Recency → (4) Coherence, because hard grammatical filters should always be applied before soft preference-based ones.

Gender and Number Agreement:

This is a hard constraint — it eliminates any candidate whose morphological features clash with the pronoun. 'He' is 3rd-person masculine singular, so only John (E1) survives; Mary, the park, and the ball are eliminated. 'She' is 3rd-person feminine singular, so only Mary (E2) survives. 'it' is 3rd-person neuter singular, so John, Mary and the dog (all animate/human) are eliminated, leaving 'a ball' and 'the park' as the only grammatically valid candidates. 'him' is masculine singular, so again only John survives among the male-consistent entities (the dog is grammatically capable of being referred to as 'it' but not standardly as 'him' in neutral register, so this filter also disfavours the dog). 'they' is plural, so only multi-entity sets survive: {John, Mary}, {John, Mary, dog}, or {John, Mary, ball} are grammatically possible; singular entities are eliminated.

Semantic Compatibility (Animacy and Action Suitability):

This constraint checks whether the candidate's semantic type is compatible with the predicate that governs the pronoun. 'He brought a ball' requires an animate, volitional agent capable of the action 'bring' — John satisfies this. 'She wanted to play with it' requires 'it' to be an inanimate, playable object; 'a ball' satisfies this perfectly while 'the park' (a location, not typically the direct object of 'play with') is semantically dispreferred, so 'it' = the ball. 'The dog chased him excitedly' requires an animate patient capable of being chased; John is a valid patient, whereas the ball is not typically 'chased' in this excited, animate sense — so 'him' = John is confirmed (and is not reflexively the dog, since the dog is the subject/agent of 'chased', not its own object). For 'they' in 'they all went home', the verb 'went home' combined with the discourse-final summarising quantifier 'all' is semantically most compatible with every animate participant who has been active in the narrative — John, Mary, and (given that dogs conventionally accompany their owners home) the dog.

Recency (Preference for the Most Recent Antecedent):

Recency is applied as a soft, ranking constraint only after the hard filters above have produced a candidate set of size ≥ 1. Since agreement and semantic compatibility already reduced each pronoun to a single strong candidate in this paragraph, recency here mainly acts as a tie-breaking confirmation: for 'it', both 'a ball' (sentence 2) and 'the park' (sentence 1) were still grammatically possible before the semantic filter; recency independently favours the more recently mentioned 'a ball', reinforcing the semantic decision. Similarly, ‘him’ in sentence 4 could in principle look back to any masculine-tagged entity, but recency strongly prefers John, the closest preceding masculine singular mention.

Coherence (Consistent Entity Chains):

Coherence is a global, discourse-level constraint: once an entity chain is established (e.g., John → He → him), every subsequent pronoun that is compatible with that chain should continue it rather than arbitrarily switching referents, unless strong local evidence forces a switch. Applying coherence confirms that ‘He’ and ‘him’ both continue the John-chain, ‘She’ continues the Mary-chain, ‘it’ continues the ball-chain, and ‘they’ continues a merged John+Mary chain (extended, by pragmatic world knowledge, to include the dog since a coherent narrative about a family outing typically returns home together with the pet).

c) Constraint Graph / Elimination Table

The resolution process can be modelled as a Constraint Satisfaction Problem (CSP): each pronoun is a variable, its domain is the set of all preceding discourse entities, and each constraint (agreement, semantic compatibility, recency, coherence) is a unary or binary constraint that prunes the domain until a unique value remains. The table below shows the variable, its initial domain, and which constraint eliminates each rejected value.

Variable	Initial Domain	Eliminated By	Final Value
He (S2)	{John, Mary, park}	Agreement removes Mary, park	John (E1)
it (S3)	{John, Mary, park, ball}	Agreement removes John, Mary; Semantic removes park	a ball (E4)
him (S4)	{John, Mary, park, ball, dog}	Agreement removes Mary, park, ball; Semantic/Coherence removes dog (agent, not patient)	John (E1)
they (S5)	{John, Mary, park, ball, dog}	Agreement removes singular entities (park, ball); Semantic removes non-participants	{John, Mary, dog}

This is naturally drawn as a constraint graph: nodes represent the pronoun-variables (He, it, him, they) and the candidate entities (John, Mary, park, ball, dog); an edge is drawn from a pronoun-node to every entity-node that survives all four filters. In this paragraph the graph collapses to a set of disjoint edges — He→John, it→ball, him→John, they→{John,Mary,dog} — which confirms the CSP is fully (and uniquely) solved with no remaining ambiguity.

d) Final Resolved Coreference Chains and Rewritten Paragraph

Coreference Chains:

- Chain 1 (John): {John, He, him, they(part)}
- Chain 2 (Mary): {Mary, She, they(part)}
- Chain 3 (the ball): {a ball, it}
- Chain 4 (the dog): {The dog, they(part)}
- Chain 5 (the park): {the park} — singleton, no further reference

Rewritten (Disambiguated) Paragraph:

“John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.”

e) Justification of Constraint Priority and Effect of Relaxing Recency

The priority order Agreement > Semantic Compatibility > Recency > Coherence was chosen because agreement is a categorical, syntax-level constraint that can be checked with zero contextual reasoning and produces the largest, cheapest reduction in the search space — it should always run first. Semantic compatibility is applied next because, unlike recency, it is still a fairly reliable, context-independent check (an inanimate object cannot literally 'bring' or be 'chased excitedly' as an agent) and further narrows the space before any probabilistic reasoning is needed. Recency is placed third because it is a statistical heuristic, not a guarantee — it is useful only for breaking ties among candidates that both agreement and semantics still allow. Coherence is placed last because it is a global, discourse-level check that is most meaningful only after local, sentence-level ambiguity has already been minimised; it acts as a final consistency validation over the whole chain rather than a per-pronoun filter.

If the recency constraint were relaxed (ignored), the resolver would have to rely purely on agreement, semantics and global coherence. In this particular paragraph the outcome would remain largely unchanged because the semantic filter already uniquely disambiguates 'it' and 'him'. However, in longer or more ambiguous discourses — for example if two grammatically and semantically compatible masculine entities existed ('John met Peter. He smiled.') — relaxing recency would leave genuine ambiguity, since the resolver would have no principled way to prefer the closer antecedent over a more distant one, forcing it to fall back on weaker coherence heuristics or to flag the mention as unresolved. This demonstrates that recency, while a soft constraint, is often the decisive factor in everyday narrative text, and its removal generally degrades resolution accuracy on pronoun-dense passages.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

a) Discourse Planning Steps Needed Before Generation

Before a natural-language response can be generated, a conversational agent must pass through several discourse-planning stages of the standard NLG pipeline. Each stage is described below as it applies specifically to this user turn ("I have an important exam tomorrow but I'm not able to concentrate.") with the required Dialog Act = Advise + Encourage.

Planning Step	Purpose	Application to this Dialog

1. Content Determination	Decide WHAT information/ideas to include	Select: acknowledge difficulty → give a concrete study/focus tip → offer encouragement about confidence
2. Discourse/Text Structuring	Order the chosen content into a coherent sequence	Cause – Effect: (cause) exam pressure → (effect) inability to concentrate; followed by Elaboration: advice that addresses the cause
3. Dialog Act Assignment	Attach the pragmatic function to each planned segment	Segment 1 = Acknowledge (empathy); Segment 2 = Advise (directive); Segment 3 = Encourage (expressive)
4. Sentence Aggregation	Combine related propositions to avoid a choppy, robotic response	Merge the advice and encouragement into 2–3 fluent sentences rather than one sentence per idea
5. Lexical Choice	Select words that satisfy the keyword and tone constraints	Choose from {focus, break, confident} plus positive-register vocabulary ('you can', 'it's okay')
6. Referring Expression Generation	Choose how to refer to entities (exam, the user) across sentences	Reuse 'exam' and 'you' consistently instead of switching to synonyms, to maintain entity coherence
7. Surface Realization	Convert the planned, aggregated content into grammatical surface text	Produce the final 2–3 sentence output, polite and encouraging in tone

b) Three Candidate Responses Satisfying All Constraints

Response 1:

"It's completely normal to feel this way before an exam — take a short break to clear your head, and then try to focus in small chunks. I'm confident you'll do better than you think."

Response 2:

"Since the exam is tomorrow, a quick break followed by focused 20-minute study sessions can really help you concentrate again. Stay confident — you've prepared more than you realise."

Response 3:

"Feeling unfocused right before an exam is common, so don't be too hard on yourself. Try to focus on one topic at a time, take a five-minute break in between, and stay confident — you've got this."

c) Evaluation of the Three Responses and Selection of the Best

Constraint	**Response 1**	**Response 2**	**Response 3**
Entity coherence (exam/concentrate/you)	Yes – reuses exam, you, focus	Yes – reuses exam, you, concentrate	Yes – reuses exam, focus, you
Discourse relation (Cause-Effect / Elaboration)	Elaboration (advice elaborates on the problem)	Cause-Effect (exam tomorrow → need to break+focus)	Cause-Effect + Elaboration (mixed)
≥ 2 keywords (focus, break, confident)	3/3 – break, focus, confident	3/3 – break, focus, confident	3/3 – focus, break, confident
Length 2–3 sentences	2 sentences – OK	2 sentences – OK	3 sentences – borderline OK
Logical consistency	Consistent	Consistent	Consistent
Politeness / positive tone	Warm, positive	Warm, positive	Warm, slightly informal ('you've got this')

Best Response Selected: Response 1.

Justification: Response 1 satisfies every constraint with the least redundancy and the tightest sentence economy (exactly 2 sentences, well within the 2–3 limit). Its discourse structure follows a clean Elaboration relation — the first clause validates the student's feeling, and the second clause elaborates with concrete, actionable advice (break → focus) before closing on an Encourage act ('confident'). It reuses the key discourse entities ('this way' referring back to the concentration problem, and 'you' consistently) without introducing new, distracting entities, which keeps the entity-coherence chain the tightest among the three. Response 2, while valid, reads slightly more mechanically because it front-loads a Cause-Effect connective ('Since…') that is less natural in spoken/typed dialog. Response 3, although fully constraint-satisfying, uses three sentences and a more casual closing phrase, making it marginally weaker on formality/coherence balance than Response 1. On a simple 0–5 coherence-and-constraint-satisfaction rubric, Response 1 scores 5/5, Response 2 scores 4.5/5, and Response 3 scores 4/5.

d) Effect of Violating Any Two Constraints

Violating Keyword Inclusion (dropping 'focus' and 'confident'):

If the response is generated without any of the required keywords ('Don't worry, everything will be fine tomorrow.'), it becomes generic and fails the Advise + Encourage dialog-act specification: it offers no actionable advice component (no reference to focusing technique) and reads as an empty reassurance rather than a targeted response to 'not able to concentrate'. This weakens task-relevance and would likely be judged unhelpful by the user.

Violating Logical Consistency:

If the system instead generated 'Don't study at all today, just relax completely, but make sure you study very hard right now,' it would contain a direct logical contradiction (relax completely vs. study very hard right now) within the same turn. This severely damages both coherence and trustworthiness — the user cannot form a single consistent action plan from contradictory advice, which is one of the most damaging failure modes for a dialog agent because it actively confuses the user rather than merely under-informing them. Together, violating keyword-inclusion and logical consistency simultaneously would produce a response that is both vague and self-contradictory — failing the dialog act entirely and reducing user trust in the system's future advice.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word "bank" and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

a) Resolving the Ambiguous Word "Bank" using Constraints

The word 'bank' has (at least) two common senses relevant here: Sense-1 (Financial institution) and Sense-2 (Land alongside a river). Word Sense Disambiguation is performed here using a constraint-driven, Lesk-style contextual overlap method: the local context window of 'bank' is compared against the gloss/typical co-occurring words of each candidate sense.

Context Clue	Compatible with Sense-1 (Financial)?	Compatible with Sense-2 (Riverbank)?
"by the river" (locative PP attached to bank)	No — financial institutions are not described as 'by the river' as a defining location in this way	Yes — riverbanks are, by definition, located beside rivers
"flooded" (predicate applied to bank)	No — a financial institution as an organisation does not 'flood'	Yes — physical land beside a river commonly floods after a storm
"after the storm"	Weak — storms rarely directly flood a bank building in a way that is the default reading	Strong — storms causing river overflow and flooding of the bank is a standard causal script
"saved by quick action"	Possible in isolation (a bank saved from bankruptcy) but not selected due to prior clues	Consistent — flood-control/rescue action saving the riverbank/nearby area

Constraint-based selection: applying Constraint 1 (Correct WSD using context) together with Constraint 4 (preserve all entities/relations — the entity must be something capable of 'flooding') and Constraint 5 (text coherence — the reading must cohere with 'by the river'), the overwhelming majority of contextual evidence eliminates Sense-1. The word 'bank' is therefore disambiguated to Sense-2: riverbank (the land bordering a river). This is a textbook case where collocational and selectional-restriction evidence (a subject of 'flood' must be a physical/geographic entity, and 'by the river' is a strong locative co-occurrence with the geographic sense) resolves the ambiguity unambiguously in favour of the geographic sense over the institutional sense.

b) Predicate Logic Representation

Let the following predicates and constants be defined:

Symbol	Meaning
bank1	the specific riverbank referred to in the sentence
river1	the river beside which bank1 lies
storm1	the storm event
flood_event1	the flooding event
save_event1	the event of bank1 being saved
riverbank(x)	x is of type riverbank
river(x)	x is of type river
location(x, y)	x is located adjacent to / at y
flood(e, x)	event e is a flooding of entity x
after(e1, e2)	event e1 occurs temporally after event e2
cause(e1, e2)	event e1 is the cause of event e2

save(e, x, m)	event e is the saving of entity x by means m
quick_action(m)	m is an instance of quick, decisive action

First-Order Predicate Logic Representation:

∃bank1 ∃river1 ∃storm1 ∃flood_event1 ∃save_event1 ∃m1 [riverbank(bank1) ∧ river(river1) ∧ location(bank1, river1) ∧ flood(flood_event1, bank1) ∧ after(flood_event1, storm1) ∧ cause(storm1, flood_event1) ∧ save(save_event1, bank1, m1) ∧ quick_action(m1) ∧ after(save_event1, flood_event1)]

In prose, this states: there exists a riverbank located at a river, a storm event that causally precedes and produces a flooding event affecting the riverbank, and a subsequent saving event that rescues the riverbank by means of an action characterised as 'quick'. Every noun phrase, verb, and the causal/temporal relation between the two clauses (linked by 'but') is captured, satisfying Constraint 4 (no information loss) and Constraint 2 (formal predicate-logic form).

c) Target-Language Sentence Satisfying All Constraints

English paraphrase (constraint-satisfying):

"Although the storm caused the riverbank to flood, prompt action saved it."

Tamil paraphrase:

"புயலையோடு ஆறுகரை வெள்ளமுட்டது, ஆனால் விரைந்த நடவடிக்கையின் காரணமாக அது காப்பாத்தப்பட்தது." (Gloss: 'Although the river overflowed with flood due to the storm, it was saved because of quick action.')

Both paraphrases keep the geographic sense of 'bank' explicit ('riverbank' / ஆறுகரை), preserve the Contrast relation using 'Although…but' / 'ஆனால்', retain both events (flood, save) and both causal participants (storm, quick action), and read as coherent, natural sentences — satisfying Constraints 3, 4 and 5 simultaneously.

d) RST-Style Discourse Tree

The two clauses of the source sentence are linked by the contrastive connective 'but', which in Rhetorical Structure Theory is analysed as a CONTRAST relation between two roughly-equal-weight spans (a multi-nuclear relation, since neither clause is purely subordinate to the other in importance — both the flooding and the rescue are equally newsworthy facts).

```
                    [CONTRAST]
                   /          \
         Nucleus 1               Nucleus 2
 "The bank by the river flooded   "but it was saved by
    after the storm"                quick action"
    (flood_event1, cause: storm1)   (save_event1, means: quick_action)
```

The Nucleus–Nucleus CONTRAST relation captures that the two situations (bad outcome: flooding; good outcome: rescue) stand in an oppositional but equally important relationship, exactly mirroring the discourse connective 'but' in the surface sentence, and satisfying Constraint 3 (maintain discourse structure).

e) Advantages of the Constraint-Based Approach over Pure Statistical MT

1. Explicit disambiguation control:

A constraint-based pipeline resolves ‘bank’ using explicit, inspectable rules (locative co-occurrence, selectional restrictions on ‘flood’). A pure statistical/neural MT system chooses a translation purely from learned probability distributions over training data, and can mistranslate polysemous words like ‘bank’ when the training corpus is skewed toward the financial sense, with no way for a user to see or correct the reasoning.

2. Guaranteed information preservation:

Because the constraint-based method builds an explicit predicate-logic meaning representation first, it can formally guarantee that every entity and relation (storm, flood, bank, quick action, causal/temporal order) is preserved in the output. Purely statistical MT optimises for fluency/likelihood and can silently drop or blend content words, especially in longer or syntactically complex sentences.

3. Discourse-structure faithfulness:

The constraint requiring the Contrast relation to be maintained ensures the target sentence keeps the same rhetorical relationship between clauses. Statistical MT systems translate largely at the sentence/segment level and have no explicit representation of rhetorical relations, so subtly shifting a Contrast into a neutral Sequence (e.g., translating ‘but’ as ‘and’) is a common, hard-to-detect error.

4. Interpretability and debuggability:

Every decision in the constraint-based pipeline (which sense was chosen, why, which predicate captures which relation) is traceable and can be audited or corrected by a linguist. Statistical/neural MT is largely a black box, making systematic errors difficult to diagnose or fix without retraining on more data.

5. Robustness on low-resource or rare constructions:

Because the constraint-based approach relies on general linguistic rules rather than learned statistics, it degrades more gracefully on sentences unlike anything in a statistical model’s training data — which is particularly valuable for a sentence containing a deliberately engineered ambiguity such as this one. The trade-off, of course, is that constraint-based systems require more manual rule/knowledge engineering and scale less easily to open-domain text than data-driven statistical MT, so in practice the two approaches are often combined (hybrid MT), using constraint-based WSD/discourse modules to guide or re-rank a statistical model’s candidate translations.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________